

Data Collection and Preprocessing Phase

Date	21-09-2026
Team ID	MCA
Project Title	Strategic Employee Attrition Intelligence dashboard
Maximum Marks	3 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Employee Dataset	Missing values in monthly_income, burnout_score, and engagement_score for a small subset of employee records	Moderate	Imputed missing numeric values using the department/job-role median in Power Query and flagged affected records for HR follow-up.
Employee Dataset	Duplicate employee_id records found for a small number of employees	High	Removed duplicate rows using Power Query's Remove Duplicates, retaining the most recently updated record per employee_id.

Employee Dataset	Inconsistent category labels in department, work_mode, and marital_status (casing/spacing mismatches, e.g. 'on-site' vs 'Onsite')	Moderate	Standardized category values using a mapping/lookup table applied during Power Query transformation.
Employee Dataset	Outlier values in commute_minutes and overtime_hours_per_week (unrealistically high entries)	Moderate	Reviewed outliers against business-realistic ranges and corrected or removed invalid entries before loading into the model.
Employee Dataset	Incorrect data types — uses_ai_tools_at_work stored as text instead of True/False, and salary_hike_pct stored as text with a % symbol	Low	Converted fields to the correct data types (Boolean and Decimal) during Power Query's Change Type step.
Employee Dataset	Invalid values in years_since_promotion and num_prior_companies (e.g. negative numbers)	High	Validated these fields against years_at_company and total_working_years, and corrected or removed records failing the logical checks.
Employee Dataset	Inconsistent labels in the attrition target column (e.g. 'Y'/'N' mixed with 'Yes'/'No')	Low	Standardized attrition values to Yes/No across all records using Power Query's Replace Values.

